MIL-C-11418G
14 August 1980
SUPERSEDING
MIL-C-11418F
12 September 1975

MILITARY SPECIFICATION

CLAMPS, LEAK REPAIR, PIPE COUPLING:

STEEL AND DUCTILE IRON,

FOR USE OVER GROOVED-TYPE COUPLINGS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers repair clamps used to provide leakproof flexible housing over leaking couplings of grooved-end pipe.

1.2 Classification. Clamps shall be the following nominal pipe sizes, as specified (see 6.2):

4-inch
6-inch
8-inch
12-inch
14-inch

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 4730

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

FF-S-85	- Screw, Cap, Slotted and Hexagon Head.
QQ-C-320	- Chromium Plating (Electrodeposited).
QQ-S-781	- Strapping, Steel, and Seals.
QQ-Z-325	- Zinc Coating, Electrodeposited, Requirements for.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-10387	- Couplings, Clamp, Pipe; with Bolts and Synthetic-Rubber Gasket for Grooved-End Pipe and Tube.

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 601	- Rubber: Sampling and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B18.2.1 - Square and Hex Bolts and Screws.
B18.2.2 - Square and Hex Nuts.
B18.10 - Track Bolts and Nuts.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 27 - Mild- to Medium-Strength Carbon-Steel Castings for General Application.
A 108 - Steel Bars, Carbon, Cold-Finished, Standard Quality.
A 183 - Heat-Treated Carbon Steel Track Bolts and Carbon Steel Nuts.
A 216 - Carbon-Steel Castings Suitable for Fusion Welding for High-Temperature Service.
A 395 - Ferritic Ductile Iron for Pressure-Retaining Castings for Use at Elevated Temperatures.
A 499 - Hot-Rolled Rail Carbon Steel Bars and Shapes.
A 575 - Merchant Quality Hot-Rolled Carbon Steel Bars.
D 395 - Rubber Property - Compression Set.
D 412 - Rubber Properties in Tension.
D 471 - Rubber Property - Effects of Liquids.
D 573 - Rubber Deterioration in an Air Oven.
D 1053 - Measuring Rubber Property - Stiffening at Low Temperature Using a Torsional Wire Apparatus.
D 1229 - Rubber Property - Compression Set at Low Temperature.
D 2240 - Rubber Property - Durometer Hardness.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Description. The clamps shall consist essentially of two body halves of steel or ductile iron as specified herein, four steel semicircular follower rings, four semicircular rubber gaskets, and an adequate number of follower bolts, body bolts, and nuts (see Figure 1 and 6.4).

3.2 First article (first produced clamp). The contractor shall furnish one or more clamps of each size as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce clamps that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

3.3.1 Steel. Steel for the clamp body shall be either a casting conforming to ASTM A 216, grade optional, or shall be fabricated by welding using carbon steel conforming to ASTM A 108, ASTM A 499, or ASTM A 575.

3.3.2 Follower rings. The semicircular follower rings shall be either a steel casting conforming to ASTM A 27, Grade 70-36, or shall be cold formed from carbon steel bar conforming to ASTM A 108, ASTM A 499, or ASTM A 575.

3.3.3 Ductile iron. Ductile iron for clamp body castings shall conform to ASTM A 395.

3.4 Width of clamp. The overall linear width of the clamp shall be not more than 8 inches for all sizes.

3.5 Weight of clamp. The weight of a completely assembled clamp shall be not more than the weight shown for each size, as follows:

Size (inches)	Weight (pounds)
4	55
6	70
8	87
12	150
14	175

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3.6 Pressure ratings. The clamps shall seal the applicable pressure specified in Table I.

TABLE I. Service pressure ratings.

Size (inches)	Working pressure (psi)	Hydrostatic pressure (psi)
4	600	1200
6	500	1000
8	450	900
12	300	600
14	300	600

3.7 Construction.

3.7.1 Clamp bodies. Clamp bodies shall be either steel or ductile iron, as specified (see 6.2). The interior of the clamp body shall be formed to permit installation of the clamp in any position over couplings conforming to MIL-C-10387, irrespective of the position of the coupling when the coupling is drawn tight over the pipe. The interior of the clamp shall provide a clearance of not less than 1/4 inch over all parts of the coupling. A 1-inch vent hole shall be drilled and tapped in the top portion of one clamp body half and fitted with a reducing bushing and 1/4-inch threaded pipe plug to permit installation of a pressure gage. Each body bolt shall be fitted with a rubber grommet to secure the bolt where the clamp body halves are separated preparatory to installation.

3.7.2 Gaskets. Gaskets shall be suitable for use with aromatic gasoline and shall be molded from a rubber compound. The gaskets shall conform to the following physical properties:

Physical requirements of the gasket

Description	Requirement
Tensile strength, initial, psi, minimum	1600
Tensile strength, oven aged 46 hours at 212° F, percent of initial, minimum	80
Tensile strength, after immersion in Reference Fuel D, ASTM D 471, for 94 hours at 73.4° F, percent of initial, minimum	45

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Physical requirements of the gasket - Continued

Description	Requirement
Tensile strength, after immersion in distilled water for 14 days at 160° F, percent of initial, minimum	65
Ultimate elongation, initial, percent, minimum	300
Ultimate elongation, oven aged 46 hours at 212° F, percent of initial, minimum	80
Ultimate elongation, after immersion in Reference Fuel D, ASTM D 471, for 94 hours at 73.4° F, percent of initial, minimum	60
Shore A hardness, initial durometer, points	65 ±5
Hardness, durometer, oven aged 46 hours at 212° F, points change, maximum	±10
Volume increase in Reference Fuel D, ASTM D 471, 94 hours at 73.4° F, percent, maximum	25
Volume change in Medium No. 4, 94 hours at 73.4° F, percent increase	zero to 15
Volume change in distilled water, 14 days at 160° F, percent increase	zero to 25
Compression set, 46 hours at 212° F, percent, maximum	40
Compression set, 94 hours at minus 20° F (fuel extracted specimens), percent, maximum	45
Torsional stiffness ratio at minus 20° F (fuel extracted specimens), maximum	5

3.7.3 Body bolts and nuts. Bolts and nuts shall conform to ASTM A 183. Bolts shall be either the oval-neck type conforming to ANSI B18.10, Class 2A fit, or square-head bolts conforming to ANSI B18.2.1, Class 2A fit, providing a bolt stop is cast in the bottom half of the sleeve to prevent the bolt from turning when the nut is tightened. Nuts shall be the heavy hexagon type conforming to ANSI B18.2.2. The bolts and nuts shall be either chromium plated in accordance with QQ-C-320, Class 2, or zinc coated in accordance with QQ-Z-325, Type I, Class 2, as specified (see 6.2). After completion of tests, the contractor shall furnish new body bolts for each clamp tested.

3.7.4 Follower bolts. Follower bolts shall conform to FF-S-85, Type II, Style 10p, Grade 5, UNF thread series Class 2A fit, 3/8-inch size. Bolts shall be plated or coated as specified in 3.7.3.

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3.8 Identification marking. The clamps shall be permanently identified in accordance with MIL-STD-130. Each body half of each clamp shall include the appropriate working pressure rating as specified in Table I, as follows: "Working pressure to _____ psi".

3.9 Treatment and painting. All parts of the clamps except the gaskets, bolts, and nuts shall be cleaned, treated, and painted in accordance with MIL-T-704, Type C, Color forest green.

3.10 Instruction card. An instruction card giving detailed installation instructions shall be provided with each clamp. The card shall be made weatherproof.

3.11 Workmanship. All parts of the clamps shall be free from defects such as inaccurately cored or drilled boltholes, burrs, sharp edges, or other defects that could impair operation or serviceability.

3.11.1 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced clamp inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

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4.3 First produced clamp inspection.

4.3.1 Examination. The first produced clamp(s) shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced clamp(s) shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.1.1 Lot. A lot shall consist of all clamps of the same size offered for inspection at one time.

4.4.2 Examination.

4.4.2.1 Individual. Each clamp shall be examined for the critical defect specified in 4.5.1. Presence of a critical defect shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 0.4 percent defective for major defects and 1.0 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.2. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The clamps shall be examined as specified herein for the following defects:

Critical

1. Holes in castings.

Major

101. Materials not as specified.
102. Dimensions not as specified.

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103. Weight of clamp not as specified.
104. Missing parts or components.
105. Construction not as specified.
106. Workmanship not as specified.

Minor

201. Identification marking missing or not legible.
202. Treatment and painting not as specified.
203. Instruction card missing or not weatherproof.

4.5.2 Tests.

4.5.2.1 Gasket. The gasket shall be tested in accordance with FED. TEST METHOD STD. No. 601 and the appropriate ASTM Test Method as specified herein. Nonconformance to the applicable requirements of 3.7.2 shall constitute failure of this test.

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Gasket physical property tests.

Description	FED. TEST METHOD STD. No. 601 Method, and ASTM Method
Tensile strength, initial	D 412
Tensile strength, after oven aging	D 573 1/
Tensile strength after immersion in Reference Fuel D	D 471 2/ 3/
Tensile strength, after immersion in distilled water (pH 7 ±0.2)	6111 3/ 9/
Ultimate elongation, initial	D 412
Ultimate elongation, after oven aging	D 573 1/
Ultimate elongation, after immersion in Reference Fuel D	D 471 2/
Hardness, initial	D 2240 4/
Hardness, after oven aging	D 573 1/ and D 2240 4/
Volume increase in Medium No. 4	6211 5/ 2/
Volume increase in Reference Fuel D	D 471 5/ 2/
Volume increase in distilled water	6211 5/ 9/
Compression set at 212° F, at 25 percent deflection	D 395 Method B 4/ 7/ 1/
Compression set at minus 20° F, at 40 percent deflection	D 1229 6/ 4/ 7/ 8/

1/ Test specimens shall be conditioned at 212° F, plus or minus 1.8° F, for 46 hours, plus or minus 1 hour.

2/ Test specimens shall be immersed in Medium No. 4, Method 6001 of FED. TEST METHOD STD. No. 601, or in Reference Fuel D, ASTM D 471, at 73.4° F, plus or minus 3.6° F, for 94 hours, plus or minus 1/4 hour.

3/ Tensile strength calculations shall be based on swollen cross-sectional area of the specimen (4.8.1 of Method 6111 of FED. TEST METHOD STD. No. 601).

4/ Test specimen shall be 1-1/8 x 7/8 x 0.500 inch, plus or minus 0.005 inch; thickness shall be attained by adding a number of plies.

5/ Test specimen shall be 2-1/8 x 7/8 inches and not over 1/16 inch thick.

6/ Test specimen shall be cut from strips that have been prepared as follows:

(a) Buff to required thickness.

(b) Immerse in Reference Fuel D, ASTM D 471, at 73.4° F, plus or minus 3.6° F, for 94 hours, plus or minus 2 hours.

(c) Blot and dry while exposed on all sides to circulating air at room temperature for 24 hours.

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(d) Dry in air-circulation oven at 130° F for 72 hours.
(e) Cool to room temperature for not less than 4 hours.

7/ Test specimens shall be compressed to the deflection indicated, regardless of hardness.

8/ Compressed specimens shall be stored at minus 20° F, plus or minus 2° F, for 94 hours, plus or minus 1/2 hour, before testing. Thickness readings shall be recorded 30 minutes after release from clamps.

9/ Test specimens shall be immersed in Medium No. 10 - distilled water (pH 7.0 ±0.2) as specified by Method 6001 of FED. TEST METHOD STD. No. 601 at 160° F for 14 days, plus or minus 1 hour.

4.5.2.1.1 Torsional stiffness ratio.

4.5.2.1.1.1 Test specimens. Test specimens shall be 0.080 inch wide x 1-1/2 inches long x 0.080 inch thick, with a 1/4-inch-square stub at each end. The specimens shall be cut from strips prepared in accordance with Note 6 of 4.5.2.1.

4.5.2.1.1.2 Test apparatus. The test apparatus shall be a torsional tester conforming to ASTM D 1053; only the torsional apparatus, the stand, a suitable bracket to hold the lower clamp, a beryllium-copper tension wire, and Type B specimens of ASTM D 1053 are required.

4.5.2.1.1.3 Test procedure. The test procedure shall be as follows:

(a) With the specimen and the test apparatus at 73° F, plus or minus 5° F, secure the lower end of the specimen in the rigidly mounted clamps of the test apparatus and secure the upper end in the upper pinch clamp attached to the lower end of the torsional wire.
(b) Adjust the pointer to the zero position by rotating the protractor scale. Turn the torsional head 180 degrees and record the pointer reading after a period of 10 seconds.
(c) After conducting the tests specified in (a) and (b) above, place the specimen in a chamber in an atmosphere of air or a mixture of air and carbon dioxide at minus 20° F, plus or minus 2° F, for 94 hours, plus or minus 1 hour. Condition the torsional apparatus and tongs for handling specimens in the chamber for not less than 2 hours before use.

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(d) Repeat the tests specified in (a) and (b) above while specimens and apparatus are still in the conditioning chamber at minus 20° F, plus or minus 2° F.

4.5.2.1.1.4 Calculations. The tests shall be conducted on three specimens from each lot sample, and the results shall be averaged. If the torsional stiffness ratio for any specimen is greater than 5, the ratio for each of the three specimens shall be within 20 percent of the averaged value of the three specimens. The torsional stiffness ratio at minus 20° F, shall be calculated by means of the following formula:

$$\text{Torsional stiffness ratio at minus } 20^\circ \text{ F} = \frac{\text{Torsional stiffness factor at minus } 20^\circ \text{ F}}{\text{Torsional stiffness factor at } 73^\circ \text{ F}}$$

$$\text{where torsional stiffness factor} = \frac{180 \text{ degrees minus angle of twist of specimen}}{\text{angle of twist of specimen}}$$

Torsional stiffness, or modulus proportionality factors may be found in Table 2 of ASTM D 1053. Average torsional stiffness ratio greater than 5 shall constitute failure of this test.

4.5.2.2 Hydrostatic. The clamp shall be subjected to the applicable hydrostatic pressure specified in Table I. The pressure shall be held for not less than 15 minutes, using water as the test fluid. Examine clamp for leakage, rupture, or yielding. Evidence of leakage, rupture, or yielding shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination.

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4.6.1.4 Samples. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

107.	Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall constitute one defect.
108.	Clamps not assembled with gaskets in place as specified for Level A.
109.	Plywood not wrapped and placed as specified for Level A.
110.	Preservation of unprotected surfaces not as specified for Level A.
111.	Attachment of instruction card to clamp for Level A not as specified.
112.	Strapping not as specified for Level A.
113.	Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. The clamps shall be assembled with the gaskets in place. A narrow strip of plywood, of sufficient thickness to relieve the gasket of pressure when the nuts are tightened, shall be placed between the body halves of the clamp and the body bolts on each end or side of the clamp. The strip of plywood shall be wrapped with barrier material conforming to MIL-B-121, Type II, Grade A, prior to installing between the body halves. The follower bolts of the assembled clamp shall be drawn fingertight. Unprotected surfaces of the body, bolts, and nuts shall be coated with Type P-1 preservative conforming to the applicable specification listed in and applied in accordance with MIL-P-116.

5.1.1.1 Instruction card. Each instruction card shall be attached to the clamp in a protected location.

5.1.2 Commercial. Each complete clamp with instruction card shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

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5.2.1 Level A. The clamps shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional, or PPP-B-621, Class 2, style optional. Blocking and bracing shall be provided to prevent free movement and damage to the gasket. Only clamps of like description shall be packed together. Strapping shall conform to QQ-S-781, Class 1 or IV, size as applicable; and, unless otherwise specified herein, shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. The clamps shall be packed as specified in 5.2.1, except that the boxes shall be Domestic Type or class as applicable. Strapping shall be Finish A.

5.2.3 Commercial. The clamps shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (Level A or Level B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The leak repair clamp is intended for use over couplings conforming to MIL-C-10387.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size clamp required (see 1.2).
(c) Time frame required for submission of first produced clamp and number of clamps required (see 3.2).
(d) Whether steel or ductile iron is required (see 3.7.1).
(e) Whether bolts and nuts shall be chromium plated or zinc coated (see 3.7.3).
(f) Degree of preservation and packing required (see 5.1 and 5.2).
(g) When other than Finish B strapping is required (see 5.2.1).

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6.3 First produced clamp. Any changes or deviations of production clamps from the approved first produced clamp during production will be subject to the approval of the contracting officer. Approval of the first produced clamp will not relieve the contractor of his obligation to furnish clamps conforming to this specification.

6.4 Information figure. Figure 1 shows a type of clamp which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of other designs which conform to this specification.

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Navy - YD

Preparing activity:
Army - ME

Project 4730-0082

Review activities:
Army - CE
Navy - MC
DLA - CS

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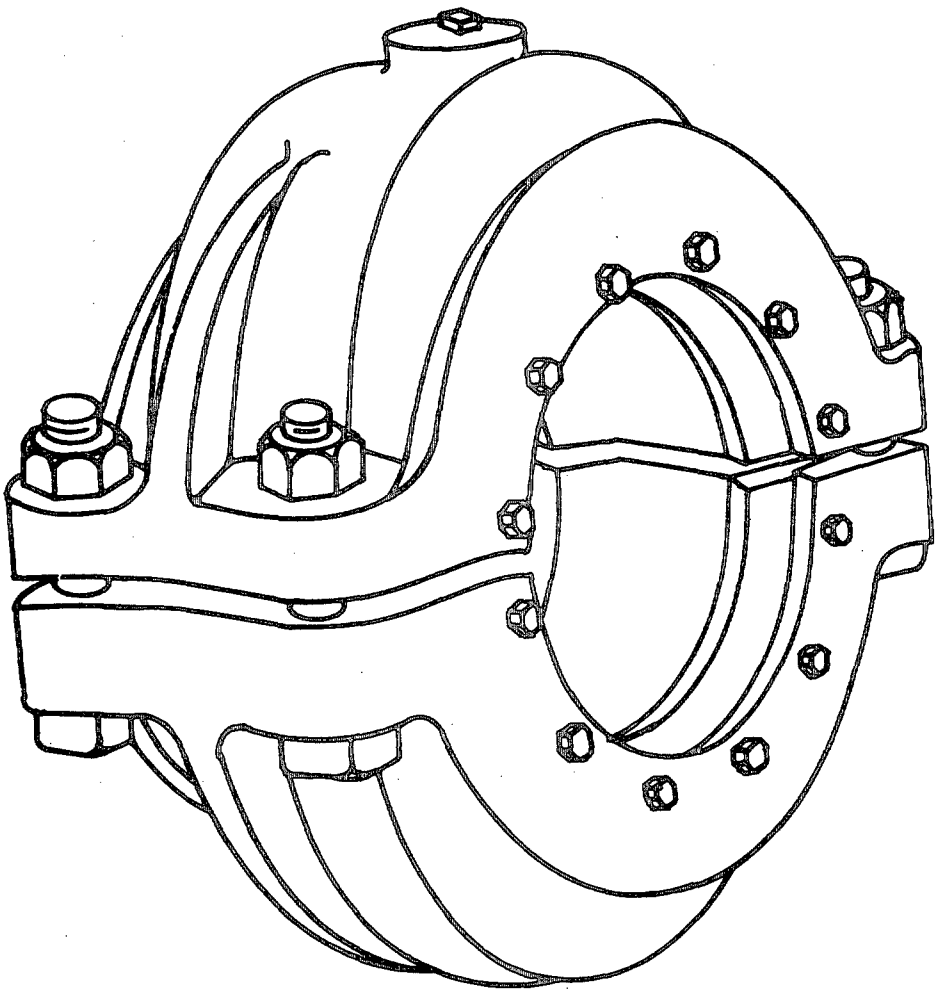


Figure 1. Over coupling leak clamp.

X1950

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